Machine Learning 2 (APM_50441_EP/DSAIB - 2026-2027). PC1

Exercise 1 (Training error optimism)

Assume that $\big((x_i, y_i)\big)_i$ is a sequence of i.i.d. random vectors. Consider a parametric model f_β fit by least squares to a set of training data $(x_1, y_1), \ldots, (x_N, y_N)$ drawn at random from a population. Let $\hat{\beta} \in \mathbb{R}^d$ be the least squares estimate, i.e.

$$\hat{\beta} = \operatorname*{argmin}_\beta \frac{1}{N} \sum_{i=1}^{N} (y_i - f_\beta(x_i))^2$$

Suppose we have some test data $(\widetilde{x}_1, \widetilde{y}_1), \ldots, (\widetilde{x}_M, \widetilde{y}_M)$ drawn at random from the same population as the training data, independently of the training data. We define

$$R_{train}(\beta) = \frac{1}{N} \sum_{i=1}^{N} (y_i - f_\beta(x_i))^2 \quad \text{and} \quad R_{test}(\beta) = \frac{1}{M} \sum_{i=1}^{M} (\widetilde{y}_i - f_\beta(\widetilde{x}_i))^2 ,$$

as well as the risk $R(\beta) = \mathbb{E}\big[(\widetilde{y} - f_\beta(\widetilde{x}))^2\big]$, where $(\widetilde{x}, \widetilde{y})$ follows the population law. For a fixed $\beta \in \mathbb{R}^d$, $R(\beta)$ is a number, while $R(\hat{\beta})$ is the random variable obtained by evaluating the function R at the random vector $\hat{\beta}$. We assume that a minimizer $\beta^\star \in \operatorname{argmin}_{\beta \in \mathbb{R}^d} R(\beta)$ exists. In what follows, the expectations are over all that is random in each expression.

1. Prove that
$$\mathbb{E}\Big[R_{train}(\hat{\beta})\Big] \leq R(\beta^\star) \leq \mathbb{E}\Big[R(\hat{\beta})\Big] .$$
Hint: start by computing $\mathbb{E}[R_{train}(\beta)]$ for a fixed $\beta \in \mathbb{R}^d$.
2. Prove that $\mathbb{E}\Big[R_{test}(\hat{\beta})\Big] = \mathbb{E}\Big[R(\hat{\beta})\Big]$.
3. Deduce that
$$\mathbb{E}\Big[R_{train}(\hat{\beta})\Big] \leq \mathbb{E}\Big[R_{test}(\hat{\beta})\Big] :$$
the training error is an optimistic estimate of the test error.

Exercise 2 (Exercise 7.4 from ESL)

Assume the additive model $Y = f(X) + \varepsilon$ where ε is centered with variance σ_ε^2, the inputs $x_1, \ldots, x_N$ being fixed. Consider the in-sample prediction error (7.18) and the training error $\overline{\mathsf{err}}$ in the case of squared-error loss:

$$\mathsf{Err}_{\mathsf{in}} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{E}(Y_i^0 - \hat{f}(x_i))^2$$

$$\overline{\mathsf{err}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{f}(x_i))^2,$$

where Y_i^0 is a new response at x_i, independent of the training responses $y = (y_1, \ldots, y_N)$. Add and subtract $f(x_i)$ in each expression and expand. Hence establish that the average optimism in the training error $\omega = \mathbb{E}_y\big(\mathsf{Err}_{\mathsf{in}} - \overline{\mathsf{err}}\big)$ is

$$\frac{2}{N} \sum_{i=1}^{N} \mathsf{Cov}(\hat{y}_i, y_i),$$

as given in (7.21).

Exercise 3 (Exercise 7.5 from ESL)

For a linear smoother $\hat{y} = Sy$, where the matrix S only depends on the fixed inputs $x_1, \ldots, x_N$, and the additive model of the previous exercise (fixed design, centered noises of variance σ_ε^2), show that

$$\sum_{i=1}^{N} \mathsf{Cov}(\hat{y}_i, y_i) = \mathsf{tr}(S)\sigma_\varepsilon^2,$$

which justifies its use as the effective number of parameters.

Exercise 4 (Hoeffding's inequality)

Let $(X_i)_{1\leqslant i\leqslant n}$ be n independent random variables such that for all $1 \leqslant i \leqslant n$, $\mathbb{P}(a_i \leqslant X_i \leqslant b_i) = 1$ where a_i, b_i are real numbers such that $a_i < b_i$. The aim of this exercise is to prove the following inequality. For all $t > 0$,

$$\mathbb{P}\left(\left|\sum_{i=1}^n X_i - \sum_{i=1}^n \mathbb{E}[X_i]\right| > t\right) \leqslant 2\exp\left(\frac{-2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right) . \tag{1}$$

1. Assume that $\mathbb{E}[X_i] = 0$ for all $1 \leqslant i \leqslant n$. Prove that it is enough to prove that for all $t > 0$,

$$\mathbb{P}\left(\sum_{i=1}^n X_i > t\right) \leqslant \exp\left(\frac{-2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right) . \tag{2}$$

2. Prove that for all $s, t > 0$,

$$\mathbb{P}\left(\sum_{i=1}^n X_i > t\right) \leqslant \mathrm{e}^{-st} \prod_{i=1}^n \mathbb{E}\left[\mathrm{e}^{sX_i}\right] . \tag{3}$$

3. Define for all $1 \leqslant i \leqslant n$, $\phi_i : s \mapsto \log\left(\mathbb{E}\left[\mathrm{e}^{sX_i}\right]\right)$. Prove that for all $s > 0$,

$$\phi_i''(s) \leqslant \left(\frac{b_i - a_i}{2}\right)^2 .$$

4. Prove that this upper bound implies for all $s, t > 0$,

$$\mathbb{P}\left(\sum_{i=1}^n X_i > t\right) \leqslant \mathrm{e}^{-st} \mathrm{e}^{s^2 \sum_{i=1}^n \frac{(b_i - a_i)^2}{8}}$$

and conclude.

Exercise 5 (Testing error and the lucky winner)

Let (X, Y) be a couple of random variables taking values in $\mathcal{X} \times \{-1, 1\}$. For a classifier $h : \mathcal{X} \to \{-1, 1\}$, the probability of misclassification is

$$L(h) = \mathbb{P}(Y \neq h(X)) ,$$

and, given independent observations $(X_i, Y_i)_{1\leqslant i\leqslant n}$ with the same distribution as (X, Y), its empirical counterpart is

$$\widehat{L}_n(h) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}_{Y_i \neq h(X_i)} .$$

We use Hoeffding's inequality proved in the previous exercise.

1. *Hold-out confidence interval.* Let h be a *fixed* classifier (for instance trained on another dataset). Prove that, for all $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$\left|\widehat{L}_n(h) - L(h)\right| \leqslant \sqrt{\frac{\log(2/\delta)}{2n}} .$$

Compute this half-width for a test set of $n = 1000$ loans and $\delta = 0.05$, and compare it with the differences of test accuracies observed in Lab 1.

2. *Empirical risk minimization.* The minimization of $\widehat{L}_n$ holds on a set $\mathcal{H}$ of classifiers, which may not contain the Bayes classifier:

$$\widehat{h}_n \in \operatorname*{argmin}_{h \in \mathcal{H}} \widehat{L}_n(h) .$$

Prove that for every set $\mathcal{H}$ of classifiers and all $n \geqslant 1$,

$$L(\widehat{h}_n) - \inf_{h \in \mathcal{H}} L(h) \leqslant 2 \sup_{h \in \mathcal{H}} \left|\widehat{L}_n(h) - L(h)\right| .$$

3. *Finite class.* Prove that when $\mathcal{H} = \{h_1, \dots, h_M\}$ for a given $M \geqslant 1$, then, for all $\delta \in (0,1)$,

$$\mathbb{P}\left(L(\widehat{h}_n) \leqslant \min_{1 \leqslant j \leqslant M} L(h_j) + \sqrt{\frac{2}{n} \log\left(\frac{2M}{\delta}\right)} \right) \geqslant 1 - \delta \, .$$

4. *The lucky winner.* In a hyperparameter search, M configurations are compared on the same validation set of size n, and the best validation error $\widehat{L}_n(\widehat{h}_n)$ is reported. Using the same tools, give a bound, valid with probability $1-\delta$, on the optimism $L(\widehat{h}_n) - \widehat{L}_n(\widehat{h}_n)$ of the reported error. Compute it for $n = 1000$, $\delta = 0.05$ and $M \in \{1, 20, 100\}$, and relate it to the experiment on the overfitting of the validation set in Lab 1 (Part 3).

Exercise 6 (Weights, proportions and costs)

We consider a binary classification problem with $Y \in \{0,1\}$ (1 for a `Risk` loan, 0 for a `No Risk` one). The training data are drawn from a law $\mathbb{P}_{\text{train}}$ in which the proportions of the classes are π_0^{train} and π_1^{train}, while the classifier will be used in the real world, with a law $\mathbb{P}$ in which the proportions are π_0 and π_1. We assume that the conditional law of X given Y is the same in both worlds (*prior shift*).

1. Let $w_y = \pi_y / \pi_y^{\text{train}}$. Prove that for any loss ℓ and any predictor f,

$$\mathbb{E}[\ell(Y, f(X))] = \mathbb{E}_{\text{train}}[w_Y \, \ell(Y, f(X))] \ .$$

2. Let $p(x) = \mathbb{P}(Y = 1 | X = x)$ and $p_{\text{train}}(x) = \mathbb{P}_{\text{train}}(Y = 1 | X = x)$. Prove that

$$\frac{p(x)}{1 - p(x)} = \frac{w_1}{w_0} \, \frac{p_{\text{train}}(x)}{1 - p_{\text{train}}(x)} \, .$$

3. Let $c(y, \hat{y})$ be the cost of predicting $\hat{y}$ when the truth is y, and assume $\Delta_0 = c(0,1) - c(0,0) > 0$ (extra cost of refusing a good loan) and $\Delta_1 = c(1,0) - c(1,1) > 0$ (extra cost of granting a defaulting one). Prove that the decision minimizing the expected real-world cost predicts 1 if and only if

$$p_{\text{train}}(x) > \tau = \frac{w_0 \Delta_0}{w_0 \Delta_0 + w_1 \Delta_1} \, .$$

4. In Lab 1, $\pi_1^{\text{train}} = 0.334$, $\pi_1 = 0.02$, and for a loan of amount a, $c(1,0) = 5 + 0.6a$, $c(0,0) = 1 - 0.05a$ and $c(0,1) = c(1,1) = 1$. Compute τ for $a = 5000$ and its limit when $a \to \infty$. Comment.

Exercise 7 (Leave-one-out for least squares)

We consider the least squares estimate $\widehat{\beta} = (\mathbb{X}^\top \mathbb{X})^{-1} \mathbb{X}^\top \mathbf{Y}$, where $\mathbb{X}$ is the $n \times d$ design matrix whose rows are the $x_i^\top$, assumed of rank d, and the *hat matrix* $H = \mathbb{X}(\mathbb{X}^\top \mathbb{X})^{-1} \mathbb{X}^\top$, of diagonal elements $h_{ii} = x_i^\top (\mathbb{X}^\top \mathbb{X})^{-1} x_i$. Let $\widehat{\beta}^{(-i)}$ be the least squares estimate computed without the observation i.

1. Prove the Sherman–Morrison formula: for an invertible matrix A and a vector u such that $1 - u^\top A^{-1} u \neq 0$,

$$(A - uu^\top)^{-1} = A^{-1} + \frac{A^{-1} u u^\top A^{-1}}{1 - u^\top A^{-1} u} \, .$$

2. Deduce that the leave-one-out residual is

$$y_i - x_i^\top \widehat{\beta}^{(-i)} = \frac{y_i - x_i^\top \widehat{\beta}}{1 - h_{ii}} \, ,$$

so that the leave-one-out error $\frac{1}{n} \sum_{i=1}^n (y_i - x_i^\top \widehat{\beta}^{(-i)})^2$ can be computed with a single fit.

3. Using $\sum_i h_{ii} = \operatorname{tr}(H) = d$, explain why the leave-one-out error is approximately the training error multiplied by $(1 - d/n)^{-2}$ when the h_{ii} are all close to d/n. Compare with the correction of Mallows' C_p seen in the lecture.

Exercise 8 (**Time series: forecasting is harder than interpolating**)
Let $(y_t)_{t\in\mathbb{Z}}$ be a stationary autoregressive process of order 1: $y_t = \phi\, y_{t-1} + \varepsilon_t$ with $|\phi| < 1$ and (ε_t) i.i.d. centered of variance σ^2.

1. Compute $\gamma(h) = \mathbb{C}\mathrm{ov}\,[y_t, y_{t+h}]$ for $h \geqslant 0$.
2. *Forecasting.* Show that the best predictor (in quadratic loss) of y_{t+h} from $(y_s)_{s\leqslant t}$ is $\phi^h y_t$, and that its mean squared error is $\sigma^2 \frac{1-\phi^{2h}}{1-\phi^2}$.
3. *Interpolating.* Show that the best linear predictor of y_t from y_{t-1} and y_{t+1} is $\frac{\phi}{1+\phi^2}(y_{t-1} + y_{t+1})$, and that its mean squared error is $\frac{\sigma^2}{1+\phi^2}$.
4. A cross validation with shuffled folds evaluates a model on observations whose neighbors in time (before *and after*) are in the training set. For $\phi = 0.9$ and a horizon $h = 12$, compare the two errors, and explain the results of Lab 1 (Part 4), and why the validation blocks must come after the training data, with a gap of at least the horizon.

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